3.

STEP BY STEP UPDATES

(Only Positive Examples)

1. Day After 3

(Positive)

Example:

(Overcast, 83, 86, False) →

Update hypothesis:

$h_1 = \langle$ Overcast, 83, 86, False $\rangle$

2. Day After 4

(Positive)

Example:

(Rain, 70, 96, False) →

Update hypothesis:

Outlook : Overcast → ?
(differs)

Temperature: 83 → ?
(differs)

Humidity : 86 → ?
(differs)

Wind : False → False
(same)

$h_2 = \langle ?, ?, ?, ?, ? \rangle$

3. Day After 5

(Positive)

Example:

(Rain, 68, 80, False) →

Update hypothesis

Outlook : ? → ? (same)

Temperature : ? → ? (same)

Humidity : ? → ? (same)

Wind : False → (same)

$h_3 = \langle ?, ?, ?, False \rangle$
(unchanged)

4. Day After 4

(Positive)

Example:

(Overcast, 64, 65, True)

Update hypothesis

Outlook ? → ? (same)

Temperature: ? → ? (same)

Humidity : ? → ? (same)

Wind : False → ? (differs)

$h_4 = \langle ?, ?, ?, ?, ? \rangle$

↓

Final Hypothesis (Maximally Specific)

$h = \langle ?, ?, ?, ?, ? \rangle$

BLOCKS IN A MACHINE LEARNING SYSTEM

1. DATA

Real world data. (images, text, numbers etc.)



2. DATA COLLECTION

Gather data from sources (sensors, files, web, etc.)



3. DATA PROCESSING

Clean, remove noise, handle missing values, normalise/scale.



4. FEATURE SELECTION/EXTRACTION

Choose important features from data.



5. TRAINING MODEL

Learns pattern from data.



6. MODEL EVALUATION

Check Performance (accuracy, precision)



7. PREDICTION/ DEVELOPMENT

Make predictions on new data / use in real world



SUPERVISED LEARNING

Uses labelled data (input + correct output) to learn and predict.

Example: Predicting whether a person has diabetes (Yes/No) based on health data.

LABELS ARE GIVEN (YES/NO)

UNSUPERVISED LEARNING

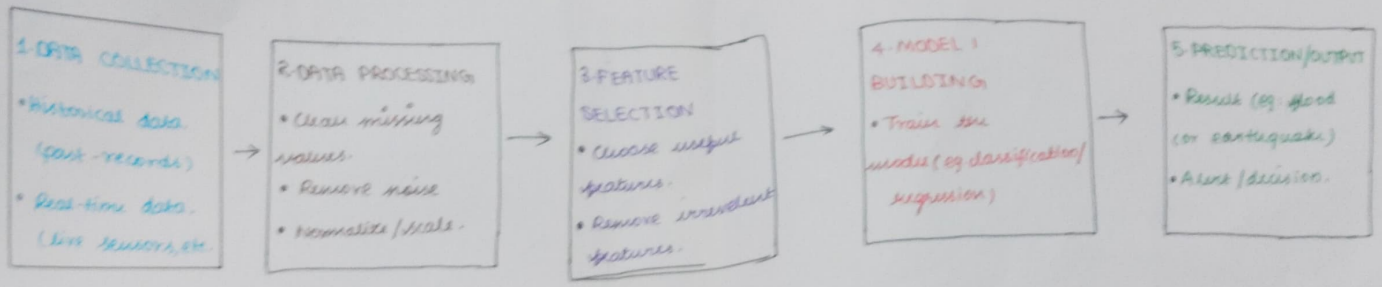
Uses unlabelled data (only input) to find hidden patterns or groups.

Example: Grouping customers into similar segments based on buying behaviour.

LABELS ARE NOT REQUIRED.

2.

MACHINE LEARNING SYSTEM - MAIN BLOCKS



TYPES OF LEARNING

SUPERVISED LEARNING

- uses labeled data (input + output)
- Eg: Predicting flood (Yes/No) using past data with labels.

Input: Rainfall, River level, moisture etc. → Output: Flood (Yes/No)

UNSUPERVISED LEARNING

- uses unlabeled data (no output)
- Eg: Grouping similar seismic data to find earthquake patterns.

Input: seismic data (no labels) → Output: clusters / groups (patterns)

4.

CANDIDATE ELIMINATION ALGORITHM

(Concept Map)

GOAL

Find the version space (sets of possible hypothesis) that classify whether a person will enjoy a sport.

ATTRIBUTES

* Weather (Sunny / Cloudy / Rainy)
* Temperature (Hot / Mild / Cool)
* Humidity (High / Normal)
* Wind (Strong / Weak)

DATASET (Training Examples)

No.	Weather	Temp.	Humidity	Wind	Enjoy Sport?
1.	Sunny	Warm	Normal	Weak	Yes
2.	Sunny	Warm	High	Strong	Yes
3.	Rainy	Cold	High	Weak	No
4.	Sunny	Cold	Normal	Weak	Yes
5.	Rainy	Warm	Normal	Strong	No.

↓

Update S and G1 after each example

1. After Example 1

S (specific boundary):

{<Sunny, Hot, High, Weak>}

G1 (General boundary):

{<?, ?, ?, ?>}

2. After Example 2:

S (specific boundary):

{<Sunny, Hot, High, ?>}

G1 (general boundary):

{<?, ?, ?, ?>}

3. After Example 3:

S (specific boundary):

{<Rainy, Cold, High, Strong>}

G1 (general boundary):

{<?, ?, ?, ?>}

4. After Example 4:

S (specific boundary):

{<~~Rainy~~, Warm, Normal, Strong>}

G1 (general boundary):

{<?, ?, ?, ?>}

Final version space

S (specific boundary)

G1 (general Boundary)

FIND-S ALGORITHM - CONCEPT MAP

GOAL

Find maximally specific hypothesis

DATASET ATTRIBUTES

Age, Income, Student, Credit Rating.

1. Start :

$S = \langle \phi, \phi, \phi, \phi \rangle$

↓

2. 1 & 2 : Negative examples → Ignore

↓

3. Positive: <Senior, Medium, No, Fair>

→ S = <Middle, High, No, Fair>

↓

4. Positive <Senior, Medium, No, Fair>

→ generalize differences

→ S = <?, ?, No, Fair>

Final Hypothesis

<?, ?, No, Fair>